PRACTICAL LAB: JAVASCRIPT OPERATORS MASTERY

Course: DFK20013 Web Design Technology

Topic: 3.1.1 Variables, Data Types, and Operators

Duration: 60 – 90 Minutes

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Date	**2/3/2026**

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1. Learning Outcomes

By the end of this lab, students should be able to:

- Implement **Arithmetic Operators** for calculations.
- Use **Assignment Operators** to update values efficiently.
- Control application states using **Increment/Decrement**.
- Validate user access using **Relational** and **Logical** operators.

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2. Practical Task: The All-in-One Operator Workshop

Create a file named lab_operators.html. Copy and complete the coding below.

HTML

```
<!DOCTYPE html>

<html lang="en">

<head>

    <meta charset="UTF-8">

    <title>DFK20013: JS Operators Workshop</title>

    <style>

        body { font-family: 'Segoe UI', sans-serif; max-width: 850px;
margin: 2rem auto; background-color: #f0f2f5; padding: 20px; color: #333;
}

        section { background: white; padding: 25px; border-radius: 12px;
margin-bottom: 25px; box-shadow: 0 4px 6px rgba(0,0,0,0.1); }

        h2 { color: #2c3e50; border-left: 5px solid #3498db; padding-
left: 10px; margin-top: 0; }

        .output-box { background: #2c3e50; color: #27ae60; padding: 15px;
border-radius: 6px; font-family: 'Courier New', monospace; margin-top:
10px; }
```

```
        .short-note { background: #fff3cd; color: #856404; padding: 15px;
border-radius: 6px; margin-top: 15px; border-left: 5px solid #ffc107;
display: none; font-size: 0.9rem; }

        input { padding: 8px; width: 100px; border-radius: 4px; border:
1px solid #ccc; }

        button { padding: 8px 20px; background: #3498db; color: white;
border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }

        button:hover { background: #2980b9; }

        code { background: #eee; padding: 2px 4px; border-radius: 3px;
font-weight: bold; color: #d63384; }

    </style>

</head>
```

```
<body>



    <header style="text-align: center; margin-bottom: 30px;">

        <h1>JavaScript Operators Mastery</h1>

        <p>Follow the steps and observe the Console and UI outputs.</p>

    </header>
```

```
    <section>

        <h2>Task 1: Basic Arithmetic (Check Console)</h2>

        <p>This task demonstrates background calculations using
<code>+</code>, <code>-</code>, <code>*</code>, <code>/</code>,
<code>**</code>, and <code>%</code>.</p>

        <button onclick="runTask1()">Run Console Logic</button>

    </section>
```

```
    <section>

        <h2>Task 2: POS System (Arithmetic & Assignment)</h2>

        <p>Price: RM 50.00. <strong>Promo:</strong> Buy > 4 units to get
RM 20.00 discount.</p>

        Quantity: <input type="number" id="qty" value="1">

        <button onclick="runTask2()">Calculate Total</button>

        <div id="outPOS" class="output-box">Final Total: RM 0.00</div>

        <div id="notePOS" class="short-note"></div>

    </section>
```

```
    <section>

        <h2>Task 3: Stock Control (Increment & Decrement)</h2>
```

```
        <p>Simulating inventory updates.</p>

        Stock Level: <span id="stockVal" style="font-size: 1.5rem; font-
weight: bold;">10</span><br><br>

        <button onclick="runTask3('inc')">Add (+)</button>

        <button onclick="runTask3('dec')"
style="background:#6c757d;">Remove (-)</button>

        <div id="noteStock" class="short-note"></div>

    </section>
```

```
    <section>

        <h2>Task 4: Gate Access (Relational & Logical)</h2>

        <p>Rules: Age 18+ OR (Age 13+ AND Parent present).</p>

        Age: <input type="number" id="age" placeholder="Age">

        <label><input type="checkbox" id="parent"> With
Parent?</label><br><br>

        <button onclick="runTask4()" style="background:#27ae60;">Verify
Entry</button>

        <div id="outSecurity" class="output-box">Status: Waiting...</div>

        <div id="noteSecurity" class="short-note"></div>

    </section>
```

Inside the <script> tag, type all the following code:

JAVASCRIPT:

1.

```
<script>
    // Helper to show explanations
    function showNote(id, text) {
        const noteDiv = document.getElementById(id);
        noteDiv.style.display = 'block';
        noteDiv.innerHTML = text;
    }
```

2.

```
// --- TASK 1: ARITHMETIC (BASIC) ---
function runTask1() {
    console.log("--- Task 1: Arithmetic Operations ---");
    let students = 31;
    students = students + 1;   // Addition
    students = students - 2;   // Subtraction
    students = students * 2;   // Multiplication
    students = students / 2;   // Division
    let power = 10 ** 2;       // Exponentiation (100)
    let remainder = 31 % 2;    // Modulus (1 - Odd)

    console.log("Final Student Count: " + students);
    console.log("10 to the power of 2: " + power);
    console.log("Remainder of 31/2: " + remainder);
    alert("Check Browser Console (F12) for results!");
}
```

3.

```
// --- TASK 2: POS SYSTEM (ARITHMETIC & ASSIGNMENT) ---
function runTask2() {
    let price = 50;
    let qty = parseInt(document.getElementById('qty').value);

    // a. Arithmetic Operator (*)
    let total = price * qty;
    let note = "<b>Subtopic (a) Arithmetic:</b> <code>*</
    code> multiplied " + price + " by " + qty + ".<br>";

    // b. Assignment Operator (-=)
    if (qty > 4) {
        total -= 20; // Shorthand for total = total - 20
        note += "<b>Subtopic (b) Assignment:</b> <code>-=</
        code> updated the total by subtracting the discount.";
    }

    document.getElementById('outPOS').textContent = "Final
    Total: RM " + total.toFixed(2);
    showNote('notePOS', note);
}
```

4.

```
// --- TASK 3: INCREMENT & DECREMENT ---
let stock = 10;
function runTask3(type) {
    let note = "";
    if (type === 'inc') {
        stock++; // (c) Increment
        note = "<b>Subtopic (c) Increment:</b> <code>++</
        code> increased the stock by 1.";
    } else {
        stock--; // (c) Decrement
        note = "<b>Subtopic (c) Decrement:</b> <code>--</
        code> decreased the stock by 1.";
    }
    document.getElementById('stockVal').textContent = stock;
    showNote('noteStock', note);
}
```

5.

```
    // --- TASK 4: RELATIONAL & LOGICAL ---
    function runTask4() {
        let age = parseInt(document.getElementById('age').value);
        let hasParent = document.getElementById('parent').checked;
        let status = document.getElementById('outSecurity');

        if (isNaN(age)) return alert("Please enter a valid age.");

        // d. Relational: >=
        // e. Logical: || (OR), && (AND)
        let isEligible = (age >= 18) || (age >= 13 && hasParent);

        status.textContent = isEligible ? "Status: ACCESS GRANTED
        ✅" : "Status: ACCESS DENIED ❌";
        status.style.color = isEligible ? "#27ae60" : "#ff4d4d";

        let note = "<b>Subtopic (d) Relational:</b> <code>>=</
        code> compared Age (" + age + ") to the limit.<br>";
        note += "<b>Subtopic (e) Logical:</b> Used <code>||</
        code> (OR) and <code>&&</code> (AND) to combine rules.";
        showNote('noteSecurity', note);
    }
</script>
```

Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

1.

Screenshot of the output

```
<script>
    // Helper to show explanations
    function showNote(id, text) {
        const noteDiv = document.getElementById(id);
        noteDiv.style.display = 'block';
        noteDiv.innerHTML = text;
    }
```

Explanation

Output does not appear as it works in html

2.

Screenshot of the output

Task 1: Basic Arithmetic (Check Console)

This task demonstrates background calculations using `+`, `-`, `*`, `/`, `**`, and `%`.

Run Console Logic

--- Task 1: Arithmetic Operations ---	LAB 2.html:114
Final Student Count: 30	LAB 2.html:124
10 to the power of 2: 100	LAB 2.html:125
Remainder of 31/2: 1	LAB 2.html:126

Explanation

IT DEMONSTRATE BASIC CALCULATION

3.

Screenshot of the output

Task 2: POS System (Arithmetic & Assignment)

Price: RM 50.00. **Promo:** Buy > 4 units to get RM 20.00 discount.

Quantity: 5 Calculate Total

Final Total: RM 230.00

Subtopic (a) Arithmetic: `*` multiplied 50 by 5.
Subtopic (b) Assignment: `-=` updated total by subtracting the discount.

	Explanation The task calculates discount rm 20 from the price rm50 whether if the quantity purchased is above 4

4.	**Screenshot of the output** Task 3: Stock Control (Increment & Decrement) Simulating inventory updates. Stock Level: **13** Add (+) Remove (-) **Subtopic (c) Increment:** ++ increased the stock by 1. Task 3: Stock Control (Increment & Decrement) Simulating inventory updates. Stock Level: **12** Add (+) Remove (-) **Subtopic (c) Decrement:** -- decreased the stock by 1.
	Explanation The tast increase number by 1 if we add and decrease number by1 if remove

5.	**Screenshot of the output**

Task 4: Gate Access (Relational & Logical)

Rules: Age 18+ OR (Age 13+ AND Parent present).

Age: 15 ☑ With Parent?

Verify Entry

```
Status: ACCESS GRANTED ✅
```

Subtopic (d) Relational: `>=` compared age (15).
Subtopic (e) Logical: Used `||` (OR) and `&&` (AND).

Explanation

3. Quick Check (Reflection)

Instruction: Answer the following questions based on your coding results.

1. **Modulus Utility:** Based on Task 1, if students % 2 returns 0, is the number even or odd?

-If students% 2 returns 0, the number is even. This is because the modulus operator returns the remainder, and an even number divided by 2 has no remainder.

2. **Shorthand Power:** In Task 2, rewrite total -= 20 using the standard assignment method.

   -The standard assignment for total -= 20 is:
   total = total - 20;

3. **Logical Priority:** In Task 4, why did we put (age >= 13 && hasParent) in parentheses?

   -We use parentheses for clarity and grouping. It ensures that the computer evaluates that specific logic (checking both age and parent status) together before comparing it with anything else. It prevents errors in the order of operations.

4. **Time Calculation:** (Extra Challenge) If a variable totalMinutes = 130, write the formula to find the remaining minutes after converting to hours using the **Modulus** operator.

5. -To find the remaining minutes after converting to hours, you use: remaining Minutes - totalMinutes % 60;
   Note: Since there are 60 minutes in an hour, 130% 60 will give you 10 (the leftover minutes).